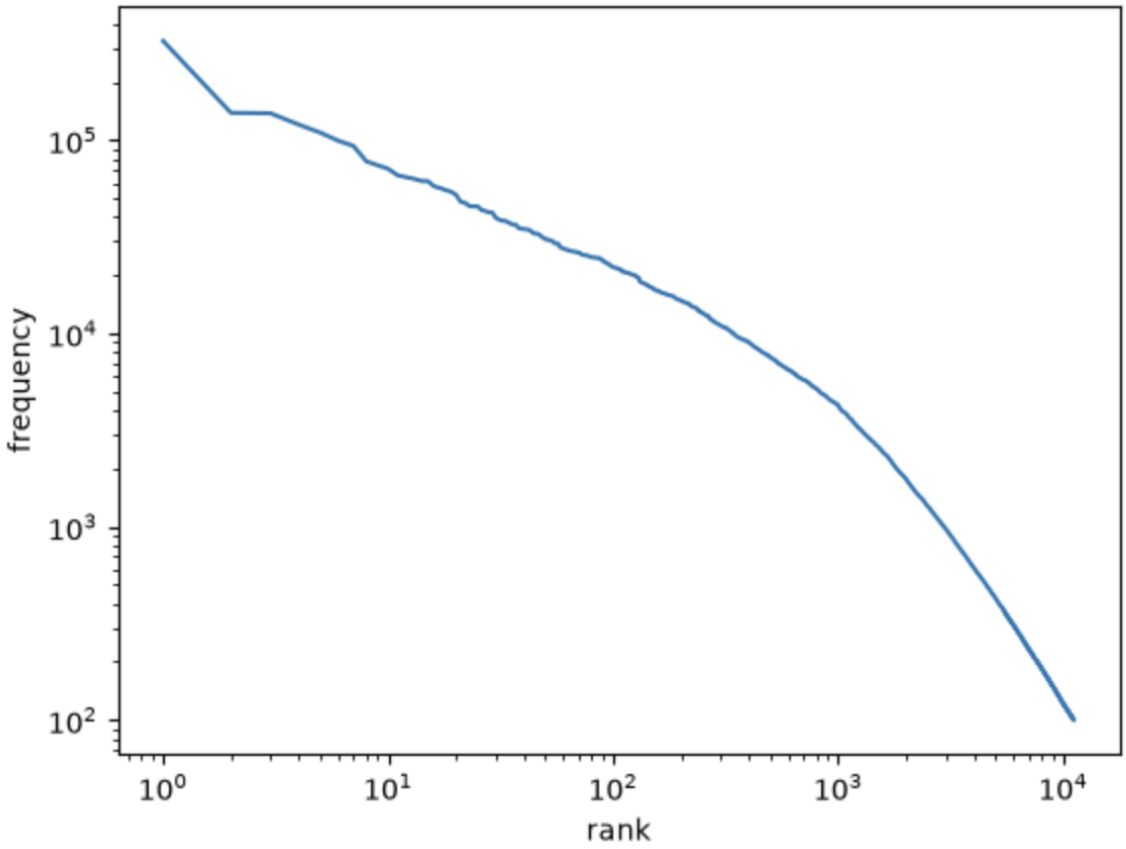
CS 6120 Assignment 2

Hoang Nguyen Bich Ngoc (Norah) - 002537880

November 3, 2026

Q1 - Plots of the distribution of words



Q2.2 - Nearest neighbors to words

0.1 Nearest neighbors of 'neural'

Top 10 words closest to neural :
neural 1.0
multiplex 0.9330724
capsule 0.9227795
regulatory 0.91713095
arnold 0.91264546
multihop 0.8922926
vehicular 0.8911211

relu 0.8882744
wireless 0.8867403
neuronal 0.88014984
Top 10 words closest to neural :
neural 0.9999999999999998
graph 0.8535556606345561
learning 0.8322024412627562
based 0.821492047716765
convolutional 0.8062276215804233
using 0.781707206615352
network 0.7704559537319637
image 0.7514999763708903
deep 0.7448076269871893
detection 0.7392665675185642

0.2 Nearest neighbors of 'machine'

Top 10 words closest to machine :
machine 1.0000001
federated 0.9904641
continual 0.9874404
imitation 0.98303425
dictionary 0.97682726
curriculum 0.9761182
meta 0.9756876
multitask 0.9752226
incremental 0.9664282
lifelong 0.9628206
Top 10 words closest to machine :
machine 1.0000000000000002
deep 0.9540344251184989
reinforcement 0.9465000707427176
based 0.8610883479528872
approach 0.8082639758024854
using 0.8049666656235361
representation 0.7920417634689754
graph 0.766321308487574
federated 0.7580851717582151
with 0.7390145452924856

0.3 Nearest neighbors of 'dark'

Top 10 words closest to dark :
dark 1.0000001
antimatter 0.9728107
baryonic 0.8726752
dense 0.8672825
particulate 0.8512486
warm 0.78892875
condensed 0.7824164
dama 0.7818416
incompressibility 0.7465162

annihilations 0.74004257
Top 10 words closest to dark :
dark 1.0000000000000002
phenomena 0.3710996325678666
thermal 0.3383886061335985
dynkin 0.32307349186474676
matter 0.32267523893655553
conception 0.32066986313198703
automorphic 0.3183925331226564
functor 0.3158847137263379
hyperon 0.31085730689927527
thresholds 0.31008278192384775

0.4 Nearest neighbors of 'string'

Top 10 words closest to string :
string 0.99999994
superstring 0.961493
gravitation 0.9519489
abjm 0.9351147
brans 0.9307954
perturbation 0.92639875
witten 0.9029094
heterotic 0.9024519
donaldson 0.90189207
iwasawa 0.9018732
Top 10 words closest to string :
string 1.0000000000000002
gauge 0.4454529754345785
effective 0.4100756204595126
finite 0.36539309540029064
dependent 0.36102541605068783
topological 0.35042021331755757
connection 0.325127006591997
gate 0.3250008589400851
scalar 0.32236665488766797
spacelike 0.3209128615512687

0.5 Nearest neighbors of 'black'

Top 10 words closest to black :
black 1.0000001
maxi 0.79979974
pockets 0.69417727
horizons 0.59063447
surrounded 0.58992034
newman 0.5760662
supermassive 0.56638575
nordstr 0.5606734
evaporation 0.5558059
binaries 0.5534988
Top 10 words closest to black :

black 1.0
chemotaxis 0.3730937915025585
gravitating 0.37098538214731724
galerkin 0.35745939025271145
bundles 0.332712186972918
terms 0.33069637357229403
ozone 0.32553360333828174
ultracold 0.3235607753941757
convective 0.32285133426352247
members 0.32180164470614975

Q3.2 - Gradient derivations

0.6 Derivation for $\partial J/\partial v_i$

$$\frac{\partial J}{\partial v_i} = \left(\sigma(v_o \cdot v_i) - 1\right) v_o + \sum_n \sigma(v_n \cdot v_i)\, v_n \tag{1}$$

0.7 Derivation for $\partial J/\partial v_o$

$$\frac{\partial J}{\partial v_o} = \Big(\sigma(v_o \cdot v_i) - 1\Big)\, v_i \tag{2}$$

0.8 Derivation for $\partial J/\partial V_n$

$$\frac{\partial J}{\partial V_n} = \boldsymbol{\sigma}(V_n v_i) \otimes v_i, \quad \forall n \tag{3}$$

Q3.4 - Plots of loss function

